

Task 5 – Social Engineering Attacks

Cyber Security Internship

Prepared as an internship task submission

1. Introduction

Social engineering is interesting from a security point of view because the attacker does not always need to break a technical control. Instead, the attacker tries to convince a person to do something that weakens security. Trust, urgency, fear and curiosity are commonly used.

2. Phishing

Phishing messages pretend to come from a legitimate organization or person. The goal may be to steal credentials, collect personal information or make the victim open a malicious link. Checking the sender and verifying unexpected requests are simple but important defenses.

3. Spear Phishing

Spear phishing is more targeted than ordinary phishing. The attacker may use information about a particular employee or organization to make the message look believable. A second verification channel is useful when a request involves money, credentials or sensitive information.

4. Vishing

Vishing is phishing performed through voice calls. A caller may pretend to be from a bank, company or technical support team. Sensitive information such as passwords, PINs and one-time passwords should not be disclosed simply because someone sounds convincing.

5. Smishing

Smishing uses SMS or similar messaging services. Messages may contain fake delivery notices, prize claims or urgent account warnings. Unexpected links should be treated carefully and verified through the organization's official website or app.

6. Baiting

Baiting uses something attractive to the victim, such as a free download or an unknown USB drive. The important lesson is not to connect unknown devices or install software simply because it is offered for free.

7. Pretexting

In pretexting, the attacker creates a believable story and identity to obtain information. Verification procedures are important because a confident or friendly person is not necessarily an authorized person.

8. Tailgating

Tailgating is a physical form of social engineering in which an unauthorized person follows an authorized person into a restricted area. Access badges, visitor procedures and refusing to hold secure doors open for unknown people can help prevent it.

9. Possible Impact

- Stolen credentials and personal information.
- Financial loss or fraudulent transactions.

- Unauthorized access to systems.
- Data breaches.
- Damage to an organization's reputation and customer trust.

10. Prevention

- Use multi-factor authentication.
- Never share passwords or one-time passwords with callers or messages.
- Verify unusual requests using a separate trusted channel.
- Provide regular security awareness training.
- Report suspicious emails, calls and messages.
- Follow physical access and visitor procedures.

11. Personal Takeaway

What stood out to me is that social engineering can look very ordinary. A message does not have to look obviously malicious to be dangerous. Taking a moment to verify a request can prevent a much larger security problem.

12. Conclusion

People are an important part of an organization's security. Good awareness, verification habits and strong technical controls together make social engineering attacks much harder to succeed.